NITIN KANOJIYA

Backend Developer | AI Engineering

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SUMMARY

Computer Engineering student at Dharmsinh Desai University with strong Data Structures and Algorithms fundamentals and hands-on experience building backend systems using Java, Spring Boot, Node.js, and Express.js. Experienced in designing REST APIs, integrating AI services, implementing asynchronous job processing, and working with relational and NoSQL databases. Built projects involving distributed processing, Docker-based code execution, Redis caching, and AI-powered code review using RAG. Seeking a Backend Development / Software Engineering Internship.

TECHNICAL SKILLS

Languages: Java, C, C++, HTML, CSS, Python, JavaScript, PHP
Frameworks: Spring Boot, Node.js, Express.js, Django, REST APIs, JWT Authentication, WebSockets (Socket.IO)
Tools: Docker, Git, GitHub, Postman, BullMQ, Judge0
Databases: MySQL, PostgreSQL, MongoDB, Redis
Concepts: REST API Design, JPA/Hibernate, OOP, JWT-based Authentication
CS Fundamentals: Data Structures & Algorithms, DBMS

EDUCATION

B.Tech, Computer Engineering — *Dharmsinh Desai University, Nadiad, Gujarat* Expected May 2028
Currently in 5th Semester | CGPA: 8.6 / 10.0 (through Semester 4)

PROJECTS

Coding Platform with Backend Build Mode | *React, Node.js, Express, MongoDB, Redis, BullMQ, Socket.IO, Docker*

- Developed a full-stack online judge with a Monaco code editor, live contests, real-time leaderboards, and instant verdicts for C++, Java, Python, JavaScript, and C submissions.
- Built a Build Mode that runs user-submitted Express.js APIs in isolated Docker containers and tests them with automated HTTP requests, using memory and CPU limits, a read-only filesystem, and network isolation.
- Designed an asynchronous job pipeline with BullMQ and Redis so code execution runs in background workers, with Socket.IO pushing results to the browser in real time.
- Integrated a self-hosted Judge0 engine with automatic failover to a cloud compiler API to keep submissions working if the primary engine is unavailable.
- Implemented JWT authentication with bcrypt password hashing, role-based admin access, and rate limiting to protect the code execution endpoint.

GitHub Repository

AI Expense Tracker | *Java, Spring Boot, React.js, PostgreSQL, JWT, REST APIs, Google Gemini AI*

- Integrated Google Gemini AI to parse raw transaction text, auto-categorize expenses, and generate personalized budget predictions.
- Enabled dynamic category creation: when an expense doesn't match an existing category, Gemini AI generates and adds a new relevant category automatically, keeping the categorization system adaptive rather than limited to a fixed list
- Implemented JWT-based authentication with Spring Security, using stateless tokens and password hashing to secure user sessions and protect account data.
- Designed a decoupled full-stack architecture, connecting a React.js frontend to a Spring Boot REST API backed by PostgreSQL for structured, relational storage of user expense data.
- Built a responsive financial dashboard in React with a deterministic color-palette algorithm to visualize live transaction data by category.

GitHub Repository

Vigilentia – AI-Powered GitHub PR Reviewer | *Node.js, TypeScript, Probot, Gemini, PostgreSQL, pgvector, Redis, BullMQ*

- Built an AI-powered GitHub App that automatically reviews Pull Requests and identifies potential bugs, security vulnerabilities, and coding-standard violations.
- Implemented RAG using Gemini embeddings and PostgreSQL pgvector to retrieve repository-specific coding guidelines for context-aware reviews.
- Designed asynchronous review processing with BullMQ and Redis, implementing atomic idempotency locks and retry mechanisms to prevent duplicate processing.

- Integrated AI-generated inline comments with changed-line validation and AES-256-GCM encryption for user-provided Gemini API keys.
  GitHub Repository

College Recommendation System | *Python, Django, MySQL*

- Designed a weighted ranking algorithm that scores and ranks colleges based on placement rate, tuition, ratings, and location, matched against student preferences.
- Built a student-facing bookmarking feature allowing users to search colleges, save favorites, and browse them by category.
- Structured a MySQL database with an admin-managed interface for maintaining listings and ensuring fast, secure search.
  GitHub Repository

ACHIEVEMENTS

- **End-to-End Application Development:** Independently designed and developed full-stack applications, covering database design, backend development, REST API integration, and frontend implementation.
- **Algorithmic Problem Solving:** Applied Data Structures and Algorithms, OOP, and problem-solving techniques to develop efficient and scalable software solutions.